**Experiment No. 09**

**Aim** : To prepare a Risk Mitigation, Monitoring, and Management Plan (RMMM) for the Construction Management System (CMS).

## Write- up :

What is Risk Mitigation, Monitoring, and Management Plan (RMMM)?
Why is RMMM important in SDLC?
What is a Risk Plan?

## Objective :

- To identify potential risks in the CMS.
- To classify risks based on **probability and impact**.
- To design mitigation and contingency plans.
- To monitor and document actions for risk handling.

### 1. Risk Identification in CMS

| Risk ID | Date | Probability (%) | Risk Level | Impact | Risk Description | Current Status |
|---|---|---|---|---|---|---|
| R1 | 06-10-2025 | 45% | Medium | High | Unauthorized access to CMS (data breach) | Open |
| R2 | 06-10-2025 | 60% | High | Medium | Incorrect task deadline entry | Open |
| R3 | 06-10-2025 | 40% | Medium | High | Project budget miscalculation | Open |
| R4 | 06-10-2025 | 50% | Medium | High | Server downtime / system crash | Open |
| R5 | 06-10-2025 | 55% | High | Medium | Delay in report generation (performance) | Open |
| R6 | 06-10-2025 | 20% | Low | Medium | User resistance to system adoption | Open |
| R7 | 06-10-2025 | 15% | Low | High | Data loss during update | Open |

## 2. Risk Mitigation Plan

| Risk ID | Mitigation Strategy |
|---|---|
| R1 | Implement multi-factor authentication, data encryption, access logs |
| R2 | Input validation, range checks, boundary value testing |
| R3 | Automatic calculation formulas, double-entry verification |
| R4 | Server redundancy, regular backups, 24x7 monitoring |
| R5 | Optimize queries, use caching, load balancing |
| R6 | Conduct user training sessions, provide user manuals |
| R7 | Regular backups before updates, rollback mechanism |

## 3. Risk Monitoring Plan

- **Security Risks (R1):** Weekly review of access logs, intrusion detection system alerts.
- **Functional Risks (R2, R3):** Automated testing reports, QA validation before deployment.
- **Technical Risks (R4, R7):** Monitoring server uptime, automated backup alerts.
- **Performance Risks (R5):** Monitoring system response time and SLA compliance.
- **Organizational Risks (R6):** Conducting feedback surveys from end-users.

## 4. Risk Management Plan

| Risk ID | Action if Occurs (Contingency Plan) |
|---|---|
| R1 | Immediately block compromised accounts, inform admin, restore from last secure backup |
| R2 | Notify user of invalid input, request re-entry, audit logs maintained |
| R3 | Financial recheck, rollback to previous budget entry, approval required |
| R4 | Switch to backup server, restore last backup data |
| R5 | Escalate to IT team, provide alternative manual reporting temporarily |
| R6 | Arrange extra training workshops, provide helpdesk support |
| R7 | Restore system from backup, re-apply updates in controlled environment |